

Queue a real 24-inch-wide demonstration

The assembled sample is **609.6 x 203.2 mm / 24 x 8 inches**. It demonstrates releasable click joints, four crossmembers and magnetic paper retention across the full roll width.

A1 Mini / PETG plate	Runs	PETG per run	Time per run
00_FIT_CHECK	1	16.0 g	47 min
01_FRAME_REPEAT_4_TIMES	4	45.2 g	118 min
02_KEYS_AND_PADS_ONCE	1	30.4 g	92 min

Frame kit: 211.3 g and about 9 h 24 min, across five manual plate runs. With fit check: 227.3 g and about 10 h 12 min, six runs. No job has been sent to a printer.

Start small: check the fit first. Then one repeat plate makes an I-shaped bay using two edge rails, one rung and two keys, with two magnetic pads. Print the other three repeats and the final keys/pads plate after that behaves well.

Main part	Quantity
DEMO_EDGE - identical edge rails with integral lugs	8
DEMO_RUNG - identical crossmembers	4
CLICK_KEY - shared connectors	14
PAPER_PAD - magnetic front pads	16

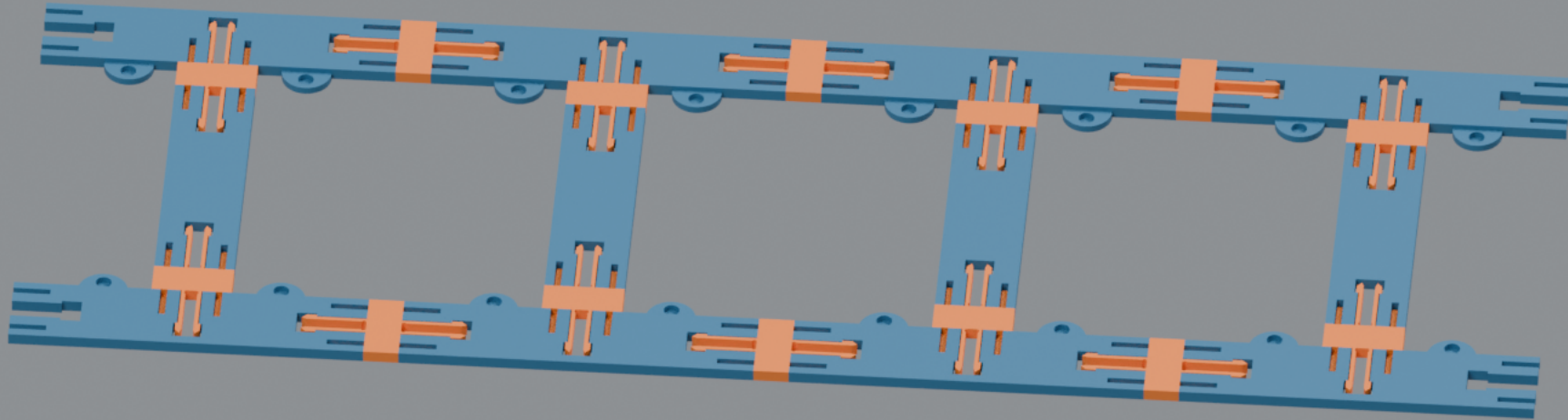
Four main types / 42 pieces. Add 32 magnets (6 x 2 mm), retaining adhesive and one 24 x 8 inch paper sheet.

Assembly and first checks

- 1. Check fit:** the final frame uses the 0.40 mm socket clearance. Rigid guides and both spring tips must engage together. Pinch the tips and withdraw straight. Keep the two fit-check receiver variants identified by their Bambu object names.
 - 2. Make two long edges:** four identical edge rails and three keys per edge. Point all D lugs toward the inside. The assembled length is $4 \times 141.9 + 3 \times 14 = 609.6$ mm.
 - 3. Connect four crossmembers:** put a key at both ends of every rung. Attach them to the four perpendicular sockets on one long edge. Work the opposite long edge onto all four free keys evenly, with its lugs facing inward.
 - 4. Attach paper:** check magnet polarity and fit before bonding. Sixteen magnets go into the integral rail holders; sixteen go into the removable pads. Let the adhesive cure. Paper lies against the smooth front face, opposite the open latches; pads go on the front of the paper.
 - 5. Observe behavior:** handle the full-width frame, check sag and joint play, peel and slide the paper gently, and compare retention with alternate pads removed. Repeatedly release a few joints and inspect the spring roots and hooks.
- Digital checks:** all supplied masters and plates fit the A1 Mini. All ten slices have supports disabled and no warnings. The assembled geometry and representative mating interfaces were checked digitally; physical fit, repeated release and magnetic grip remain untested.
- Scope:** this is a ladder-shaped demonstration, not a complete gate border or a wind-rated assembly. It has no dedicated PVC-dock interface. The sacrificial stacks and PVC dock remain separate optional experiments.

24-inch paper frame / actual printable parts

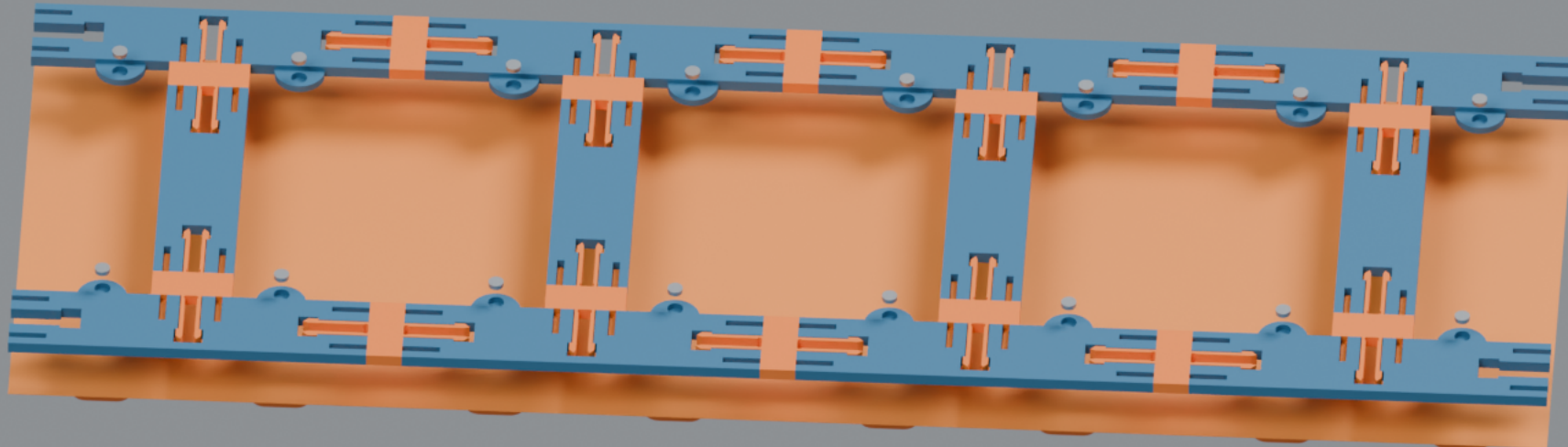
609.6 x 203.2 mm. Eight identical edge rails, four crossmembers and fourteen replaceable click keys. All joints release from the rear.



Four main printed types including the front pads. This ladder specimen tests click fit, full-width handling and paper grip; it is not a wind-rated gate border.

Cut one 8-inch length from the 24-inch roll

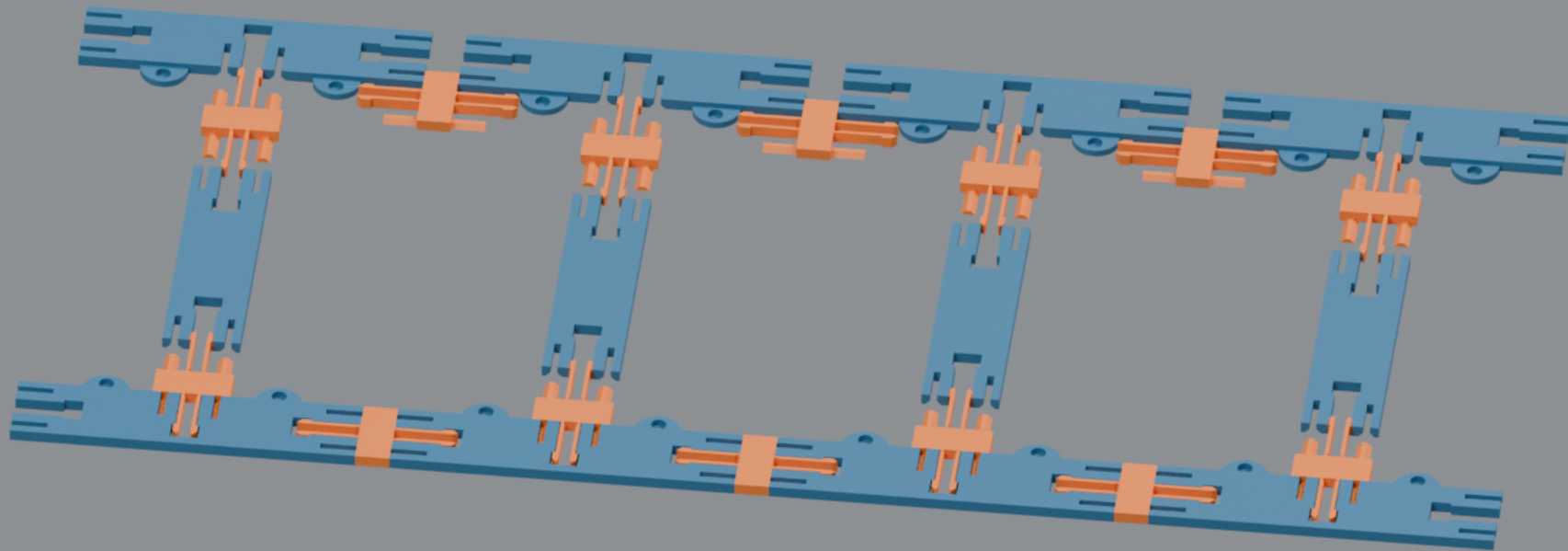
Thirty-two 6 x 2 mm magnets form sixteen pairs. Integral D-shaped lugs carry the rear magnets; front pads spread pressure on the paper.



Exploded vertically. Assemble paper against the smooth front face ($z=0$). Check polarity and magnet fit before bonding magnets into their pockets.

Two long edges, then four crossmembers

Build each 609.6 mm edge with four rails and three keys. Add a key to each end of each crossmember, then close the second edge across all four.



Keep the D lugs facing inward and all pockets on the rear. Engage the rigid guides straight; inspect both hooks. Fit-check first before printing four repeat plates.

Print this plate four times

Each run: two edge rails, one crossmember, two keys and two front pads. A separate single plate supplies the remaining six keys and eight pads.

